Table of Contents

iMotions Lab > Sensor Setup > Shimmer (GSR, ECG, EMG, EXG)

Shimmer 3 GSR+: Device Connection and Configuration in iMotions 2

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

Shimmer 3 GSR+: Device Connection and Configuration in iMotions

With the Shimmer GSR+ device you can collect the following biophysiological signals:

- Galvanic Skin Response (GSR), also referred to as Electrodermal Activity (EDA) or Skin Conductance (SC).
- Optical Heart Rate, also referred to as Photoplethysmatogram (PPG).

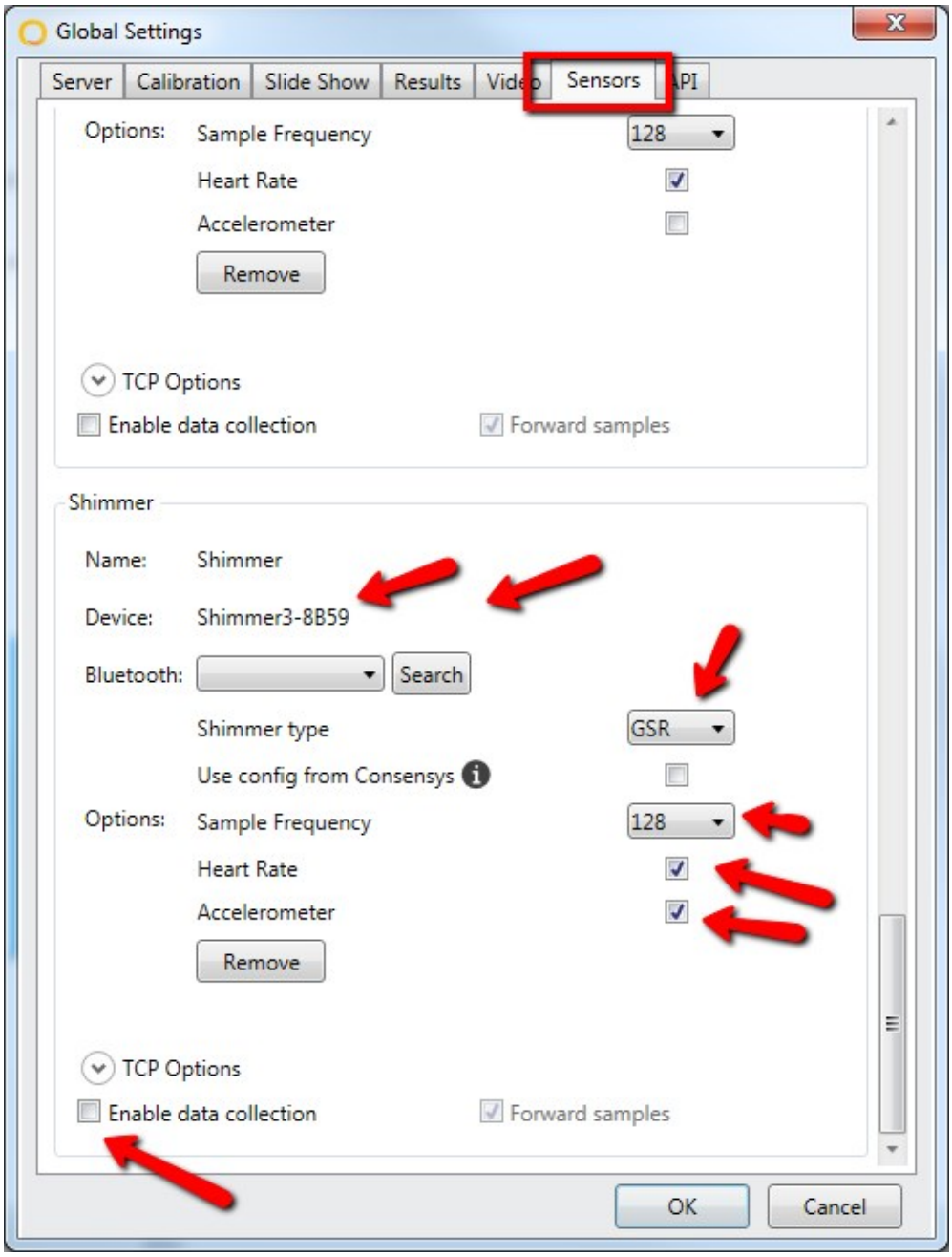
Additionally, the Shimmer GSR+ device has integrated several telemetry sensors such as acceleration, gyro, magnetometer, and pressure sensors. More details can be found target="_blank" rel="noopener noreferrer">Shimmer website or in the PDF below.



The connection and configuration of the Shimmer GSR+ sensor is accomplished directly in iMotions:

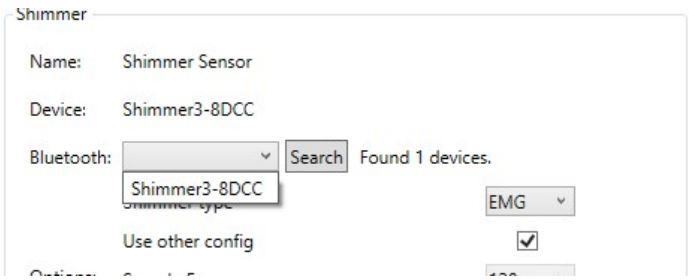
1. Connect the Shimmer GSR+ sensor to the computer's Bluetooth (this can be completed on Windows 10 by going to "Bluetooth & Devices", clicking "Add device", and then clicking on "Bluetooth". Select the Shimmer device from the list). If you are asked for a pin, enter "1234".
2. In iMotions, click “Preferences” > “Global Settings”, and select the tab “Sensors” [A]. Identify the Shimmer GSR device section by scrolling down.

Confidential © 2023 Redistribution is not permitted without written permission from iMotions



3. First make sure your Shimmer device is on, charged and configured (For help see:)
4. Click "Search" next to the Bluetooth section. iMotions will find all the Shimmer devices within range.
5. Use the pull down menu to select the correct device you want to pair with. The device ID can be found on the back of the device as four alphanumeric characters.

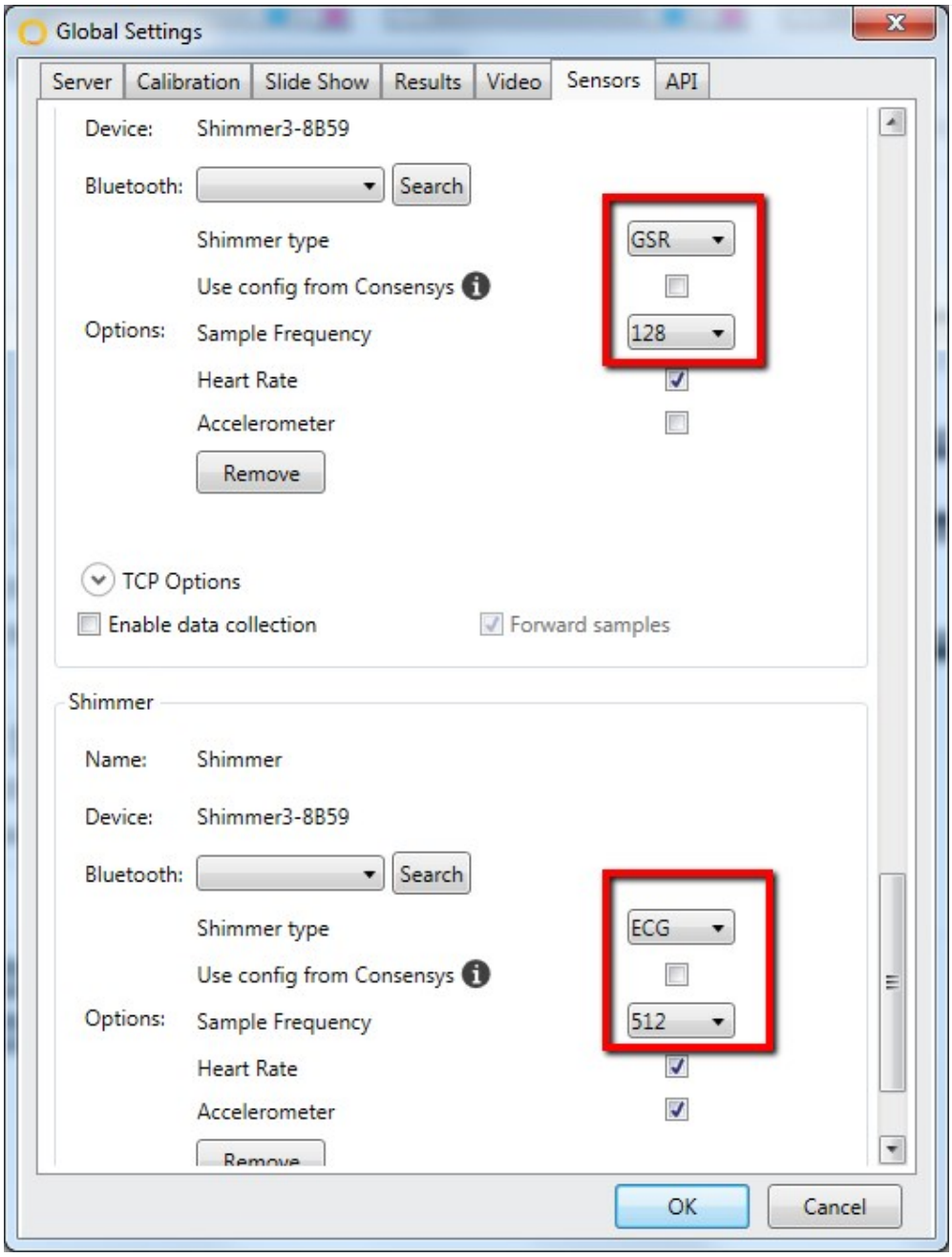
Confidential © 2023 Redistribution is not permitted without written permission from iMotions



6. Select "GSR" under Shimmer Type.
7. Uncheck "Use Config from Consensys". This box should only be checked off if you wish to use the default Shimmer configuration as defined by Consensys, which will override iMotions Global Settings. When this box is unchecked, the Options parameters will become available for modification (see below). Hover over the 'i' for more information.



Confidential © 2023 Redistribution is not permitted without written permission from iMotions



8. Options: The Options parameters will become available by un-checking the "Use other config" box.



The parameters are described below:

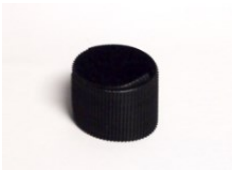
- Sample Frequency: For GSR, a sampling rate of 128 is sufficient. Theoretically, GSR can be collected at sampling rates lower than this, but the lowest sampling rate available with iMotions is 128, to allow for low-pass filtering of the data. Thus, if you wish to use a lower sampling rate, either: 1) downsample the raw data after export from iMotions, 2) checking off "use other config" and manually entering the new sampling

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

rate in Consensys.

- Heart Rate: Check this box if you intend to also collect PPG with your collection. You will need the PPG earpiece or finger cuff, depending on what hardware has come with your Shimmer kit. Please refer here for more information on PPG.

PPG Ear Piece:  Finger cuff:



- Accelerometer: Check this box if you want acceleration in the X,Y, and Z planes to be included in your data collection.

Finally, enable the check-box "Enable data collection", and press OK.

The Shimmer widget in the LiveViewer should display 'Connecting', followed by 'Configuration' and 'Collecting'.

Powered by Document360

Confidential © 2023 Redistribution is not permitted without written permission from iMotions

Confidential © 2023 Redistribution is not permitted without written permission from iMotions